

INTERNSHIP PROJECT REPORT

SAP Procure-to-Pay (P2P) Risk Analytics

Business Requirement Documentation, Synthetic SAP Datasets, Excel-based Analysis & Power BI Dashboard Development

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Acknowledgement

I would like to express my sincere gratitude to my mentors and the entire team for their continuous guidance, patience and encouragement throughout the course of this internship. Their willingness to explain SAP procurement concepts, review my Business Requirement Documents, and critique early versions of the dashboards was instrumental in shaping the quality of this project.

I am also thankful to the organization for providing me the opportunity to work on a live-style analytics engagement, exposing me to how procurement risk is analysed at a professional consulting level, from requirement gathering through to executive dashboard delivery. This experience has given me a practical understanding of how technology, business logic and data analytics come together to support real decision-making in a Procure-to-Pay environment.

Finally, I extend my appreciation to my peers and family for their constant support and motivation during this internship.

Summary

This report documents a 1.5-month internship engagement focused on SAP Procure-to-Pay (P2P) Risk Analytics. The objective of the internship was to simulate, end-to-end, how a risk analytics function inside a large organization identifies procurement exceptions, quantifies their financial impact, and presents them to management through interactive dashboards. The engagement was structured around six distinct risk scenarios spanning purchase price governance, PO tolerance control, vendor payment term consistency, vendor master data integrity, duplicate payment exposure, and three-way match compliance.

The project followed a disciplined, consulting-style methodology. Each risk area began with a formal Business Requirement Document (BRD) that articulated the objective, the underlying business risk, the financial or compliance impact, the relevant SAP tables and fields, and the precise business logic used to isolate exceptions. Because live SAP production data could not be used for confidentiality reasons, structurally realistic synthetic datasets were generated in Excel to mirror SAP purchasing, vendor master, and financial accounting tables such as EKKO, EKPO, EKBE, LFA1, LFB1, BSIK and BSAK. Every scenario was then independently validated using Pivot Tables before any dashboard was built, ensuring that the business logic produced defensible, auditable exception counts and financial figures.

The validated datasets were imported into Power BI, where Power Query was used for data shaping and DAX measures were authored to calculate KPIs such as total financial impact, exception counts, and compliance percentages. Six interactive dashboard pages were designed which were:

- Purchase Price Variance Analysis
- PO Tolerance Variance Analysis
- Multiple Payment Terms Analysis
- Vendor Master Data Monitoring
- Duplicate Payment Risk Analytics
- Three-Way Match Compliance Analytics

Each incorporating KPI cards, decomposition trees, donut and bar visualizations, cross-filtering, and drill through pages that allow a user to move from a summary figure down to the individual purchase order or invoice driving it.

This report captures the complete journey: the P2P process context, the methodology followed, a detailed walkthrough of each of the six risk scenarios with their KPIs and visuals, the dashboard's design features, consolidated findings, recommendations, and the key learning outcomes from the internship. The intent throughout has been to demonstrate not just technical dashboard-building ability, but the analytical judgment expected of a risk and controls professional in a Procure-to-Pay environment.

Chapter 1: About the Company

Ernst & Young (EY) is one of the world's largest professional services organizations and a member of the globally recognized Big Four accounting and consulting firms. Operating in more than 150 countries, EY provides a broad range of services including Assurance, Consulting, Strategy and Transactions, and Tax. The firm helps organizations improve business performance, manage risks, embrace digital transformation, and create long-term sustainable value.

A key area of EY's Risk Consulting practice focuses on Procure-to-Pay (P2P) analytics, where transactional data from SAP systems is analysed to identify operational inefficiencies, policy violations, compliance gaps, and potential fraud risks. By leveraging business intelligence tools such as Microsoft Power BI, EY enables organizations to shift from traditional manual audits to continuous monitoring through automated dashboards and exception reporting.

During this internship, exposure was gained to the end-to-end lifecycle of a consulting engagement involving SAP P2P risk analytics. This included understanding procurement business processes, identifying key risk indicators (KRIs), preparing Business Requirement Documents (BRDs), performing data analysis using Microsoft Excel, developing interactive Power BI dashboards, and translating business requirements into meaningful visual insights for decision-making. The project emphasized how data analytics can strengthen procurement controls, improve compliance, and support proactive risk management in enterprise environments.

Chapter 2: Procure-to-Pay (P2P) Process

Procure-to-Pay, commonly abbreviated as P2P, is the end-to-end process by which an organization identifies a need for goods or services, procures them from an approved vendor, receives and verifies them, and finally settles payment. Because P2P sits at the intersection of operations, finance and vendor management, it is one of the most closely monitored processes in any large enterprise. A small breakdown in control at any stage can translate directly into financial leakage, compliance exposure, or strained vendor relationships. Understanding this lifecycle in detail was the foundation on which every risk scenario in this project was built.

2.1 The P2P Lifecycle

Identify the need → Purchase Requisition (PR) → Request for Quotation (RFQ) → Supplier Selection → Purchase Order (PO) → Order Fulfilment → Goods Received Note (GRN) → Invoice Received → Invoice Processing → Accounts Payable

2.2 Why Internal Controls Matter

Each of the stages above is a control point, not merely a processing step. A well-controlled P2P process ensures that spend is authorized before it is committed, that pricing is consistent and competitively benchmarked, that goods and invoices are independently verified before payment, and that vendor data is accurate and unique. When these controls weaken, whether through manual overrides, master data errors, or gaps in system configuration, the organization is exposed on multiple fronts simultaneously: direct financial leakage through overpayment or duplicate payment, audit findings and regulatory penalties from non-compliant tax

documentation, and reputational or relationship damage with vendors and regulators. Continuous, dashboard-based monitoring of the kind built in this project exists precisely to catch these breakdowns early, well before they surface in a formal audit.

2.3 SAP Tables Used in This Project

Every scenario in this project was designed against real SAP table and field structures, even though the underlying data was synthetically generated. Working with authentic table names and relationships was deliberate and it ensured that the business logic, joins and filters built in Excel would translate directly to a live SAP environment. The tables referenced across the six scenarios are described below.

EKKO — Purchasing Document Header

EKKO stores one record per purchase order at the header level, holding fields such as EBELN (PO Number), LIFNR (Vendor Number), BEDAT (PO Date) and BUKRS (Company Code). It is the anchor table for any PO-level analysis because it identifies who the PO was raised for, by which company code, and on what date — information that is essential when grouping purchase orders to detect price or vendor inconsistencies.

EKPO — Purchasing Document Item

EKPO holds the line-item detail beneath each EKKO header — MATNR (Material Number), WERKS (Plant), NETPR (Net Price), MENGE (Quantity), BPRME (Order Unit) and tolerance fields such as UEBTO and UNTTO. Because pricing, quantity and tolerance all live at the item level, EKPO was the primary working table for the Purchase Price Variance and PO Tolerance scenarios.

EKBE — History per Purchasing Document

EKBE records the transactional history against a PO — goods receipts and invoice receipts — distinguished by the VGABE (transaction/event type) field. This table made it possible to reconstruct, for each PO line, what quantity was received (VGABE = 1) versus what quantity was actually invoiced (VGABE = 2), forming the backbone of the Three-Way Match Compliance scenario.

EKET — Scheduling Agreement Schedule Lines

EKET captures delivery schedule information for a PO line, including planned delivery dates and quantities. While not the primary table for any single scenario in this phase of the project, it was reviewed as part of understanding delivery-driven exceptions such as late or partial GRNs, which are referenced as possible causes behind several of the flagged exceptions.

EBAN — Purchase Requisition

EBAN holds the purchase requisition that precedes a PO, capturing the originating business need, requested quantity and requisitioner. It provided useful context for understanding how urgent or ad-hoc requisitions can, further downstream, manifest as price variance or tolerance exceptions.

LFA1 — Vendor Master (General Data)

LFA1 is the client-independent vendor master table, storing NAME1 (Vendor Name), STCD1 (PAN Number), STCEG (GST Number) and other core identifiers. This table was central to the Vendor Master Validation scenario, where PAN and GST formats were checked and duplicate identifiers across vendor codes were identified.

LFB1 — Vendor Master (Company Code Data)

LFB1 extends LFA1 with company-code-specific vendor data such as payment terms and reconciliation account assignment. It was joined with LFA1 throughout the vendor master checks to give company-code context to PAN and GST exceptions.

BKPF — Accounting Document Header

BKPF stores the header of every financial accounting document — document number, posting date and company code. It provides the umbrella record under which individual line items in BSEG sit, and was used conceptually when tracing invoice postings back to their accounting documents.

BSEG — Accounting Document Segment

BSEG holds the detailed line items of every accounting document, including amounts and posting keys. Together with BKPF, it represents the general ledger view of a transaction, complementing the vendor sub-ledger tables used for duplicate payment detection.

BSIK — Vendor Line Items (Open)

BSIK contains vendor line items that are still open (not yet cleared) — including LIFNR, XBLNR (Vendor Invoice Number), BUDAT (Posting Date) and WRBTR (Amount). It was one of the two source tables used in the Duplicate Payment Risk scenario to identify invoices awaiting payment.

BSAK — Vendor Line Items (Cleared)

BSAK mirrors BSIK but for line items that have already been cleared (paid). Combining BSIK and BSAK gave a complete view of vendor invoices across their lifecycle, which was essential for detecting duplicate invoice numbers, dates and amounts irrespective of whether the duplicate had already been paid.

MARA — General Material Master

MARA stores the client-independent material master, including MATNR and base material attributes. It provided the material dimension referenced throughout the Purchase Price Variance and PO Tolerance scenarios.

MAKT — Material Descriptions

MAKT holds language-specific text descriptions for each material number in MARA. While the synthetic dataset used simplified material codes (MAT100, MAT101, etc.), MAKT was reviewed to understand how material descriptions would enrich exception reporting in a live environment, making dashboards more readable for business users unfamiliar with raw material codes.

Chapter 3: Project Methodology

The project was executed in four disciplined stages

- Business Requirement Documentation
- Dataset Creation
- Pivot Table Analysis
- Power BI Dashboard Development

Mirroring how a professional risk analytics engagement would be run inside a consulting or internal audit function. This staged approach ensured that every dashboard number could be traced back to a documented business rule, rather than being built ad hoc.

3.1 Business Requirement Document (BRD)

Every risk scenario was first documented in a **Business Requirement Document (BRD)** before any data creation began. Each BRD defined the business objective, associated risk, impact, required SAP tables and fields, and the business logic for identifying exceptions. This ensured clear requirements, prevented scope changes, and established a strong foundation for dataset creation and dashboard development.

3.2 Dataset Creation

As real SAP production data could not be used due to confidentiality, **synthetic datasets** were created in Excel following SAP table structures and naming conventions. The data was intentionally designed to include realistic exception scenarios such as price variations, duplicate invoices, tolerance inconsistencies, and three-way match mismatches. This ensured meaningful analysis while maintaining complete data privacy.

3.3 Pivot Table Validation

Before building the dashboards, each scenario was validated using **Excel Pivot Tables** to verify that the business logic produced the expected exceptions. Key metrics such as counts, minimum and maximum values, and financial impact were analysed and cross-checked. This validation ensured that all Power BI KPIs and visuals were based on accurate and reliable data.

3.4 Power BI Dashboard Development

The validated datasets were imported into **Power BI**, where Power Query was used for data preparation and DAX measures were created for dynamic KPIs. Interactive dashboards were developed using KPI cards, charts, decomposition trees, slicers, tooltips, and drill through functionality. These features allow users to analyse procurement risks, investigate exceptions, and gain actionable insights through an intuitive interface.

Chapter 4: Risk Scenarios

Scenario 1 — Purchase Price Variance

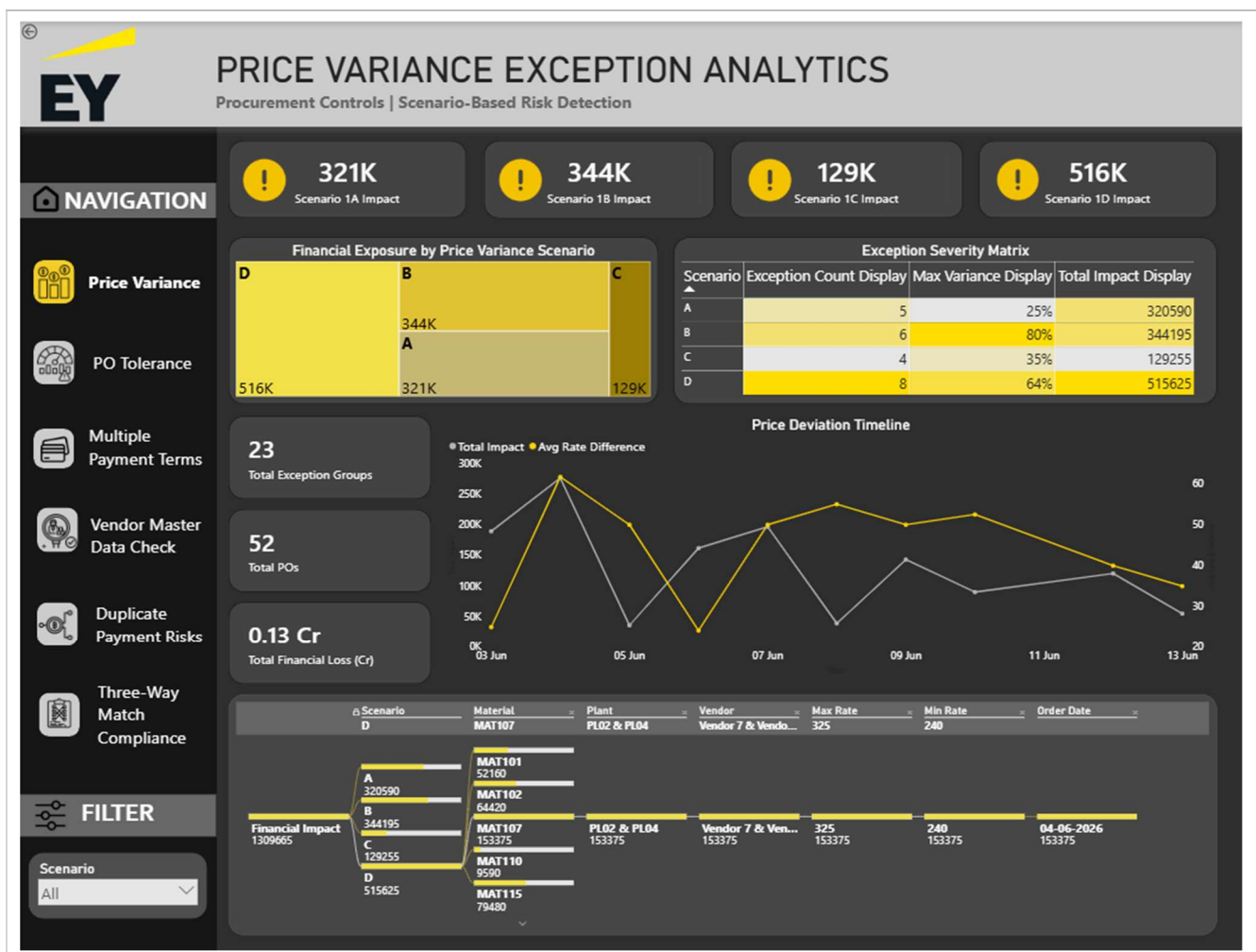
Objective

This scenario evaluates whether identical materials are being procured at consistent prices and identifies instances of purchase price variance. Four sub-scenarios were analysed: price differences for the same vendor and plant, across different vendors, across different plants, and across both vendors and plants. Together, these analyses help identify financial leakage, weak procurement controls, inconsistent vendor negotiations, and opportunities to standardize pricing across the organization.

Business Risk

Price inconsistencies can lead to unnecessary procurement costs, weaken centralized purchasing controls, and create audit concerns due to unexplained rate variations. They also reduce the organization's bargaining power with vendors and make budgeting and procurement forecasting less reliable.

Dashboard Overview



Purchase Price Variance Analysis Dashboard

KPI Explanation

Scenario A / B / C / D Financial Impact: These KPI cards display the financial exposure for each price variance scenario, helping identify where the highest procurement risk exists. Scenario D contributes the highest financial impact, indicating that price inconsistencies are most significant across different plants and vendors.

Total Exception Groups (23): Represents the number of unique pricing exception groups identified based on the defined business logic, highlighting the number of distinct pricing issues rather than individual transactions.

Total Purchase Orders (52): Shows the total number of purchase orders involved in all identified price variance exceptions, indicating the overall scale of affected transactions.

Total Financial Impact (₹13.09 Lakhs): Displays the combined monetary impact of all price variance exceptions, providing management with an estimate of the potential financial exposure.

Dashboard Visuals Explanation

Financial Impact by Price Variance Scenario: This visual compares the financial impact across all four scenarios, making it easy to identify which procurement pattern contributes most to overall exposure. It supports quick prioritization of investigation efforts.

Financial Impact vs. Rate Variance Trend: This trend chart compares financial impact with average rate variance over time. It helps identify dates where unusual pricing behaviour occurred and highlights periods that may require further procurement review.

Financial Impact Breakdown (Decomposition Tree): The decomposition tree allows users to drill down from the overall financial impact into scenario, material, plant, vendor, and purchase order details. It provides an interactive way to trace the root cause of pricing exceptions.

KPI Cards: The KPI cards update dynamically based on slicer selections, allowing users to instantly view the financial impact, affected purchase orders, and exception groups for any selected scenario

Business Insights

The analysis indicates that purchase price inconsistencies are concentrated within specific materials and vendor combinations rather than being uniformly distributed. The highest financial exposure arises from **Scenario D**, suggesting that decentralized procurement across different plants and vendors is the primary driver of pricing variation. Additionally, significant price differences observed in Scenario B indicate opportunities to strengthen vendor benchmarking and negotiate more consistent procurement rates across the organization.

Possible Real-World Causes

- Urgent or emergency procurement at one plant forcing acceptance of a premium rate to avoid a production stoppage
- Raw material or commodity price inflation between two nearby order dates that were still grouped as "same date" comparisons at month boundaries
- Vendor-specific volume discounts where one plant ordered a larger quantity and legitimately received a lower unit rate
- Freight, transportation or logistics cost differences embedded in the net price for plants located further from the vendor
- Exchange rate fluctuation where a portion of the material cost is import-linked
- Multiple plants historically negotiating independently with the same vendor without a shared rate card
- ERP configuration allowing manual price overrides at the PO line level without a secondary approval
- Vendor master data issues where two vendor codes actually represent the same legal vendor, splitting volume and pricing history
- Human error during manual PO creation, particularly copy-paste from an outdated PO template
- Tax or duty structure differences across plant locations affecting the net price field
- Genuine vendor negotiation outcome where a newer contract rate had not yet been applied uniformly across all open POs
- Rush or partial-delivery premiums charged for expedited fulfilment on select orders

Recommended Controls

- Implement a centralized rate-card / contract-price table in SAP so that NETPR is validated against an approved price list at PO creation
- Introduce a system-enforced approval workflow for any PO price exceeding the approved rate-card value beyond a defined tolerance
- Consolidate vendor negotiations centrally for materials procured across multiple plants to eliminate fragmented pricing
- Run a periodic (weekly or monthly) automated price-variance exception report rather than relying solely on year-end audit sampling
- Strengthen vendor master governance to eliminate duplicate vendor codes that fragment purchase history and pricing visibility
- Require a documented business justification whenever a manual price override is applied at the PO line level

Scenario 2 — Tolerance Limit Monitoring

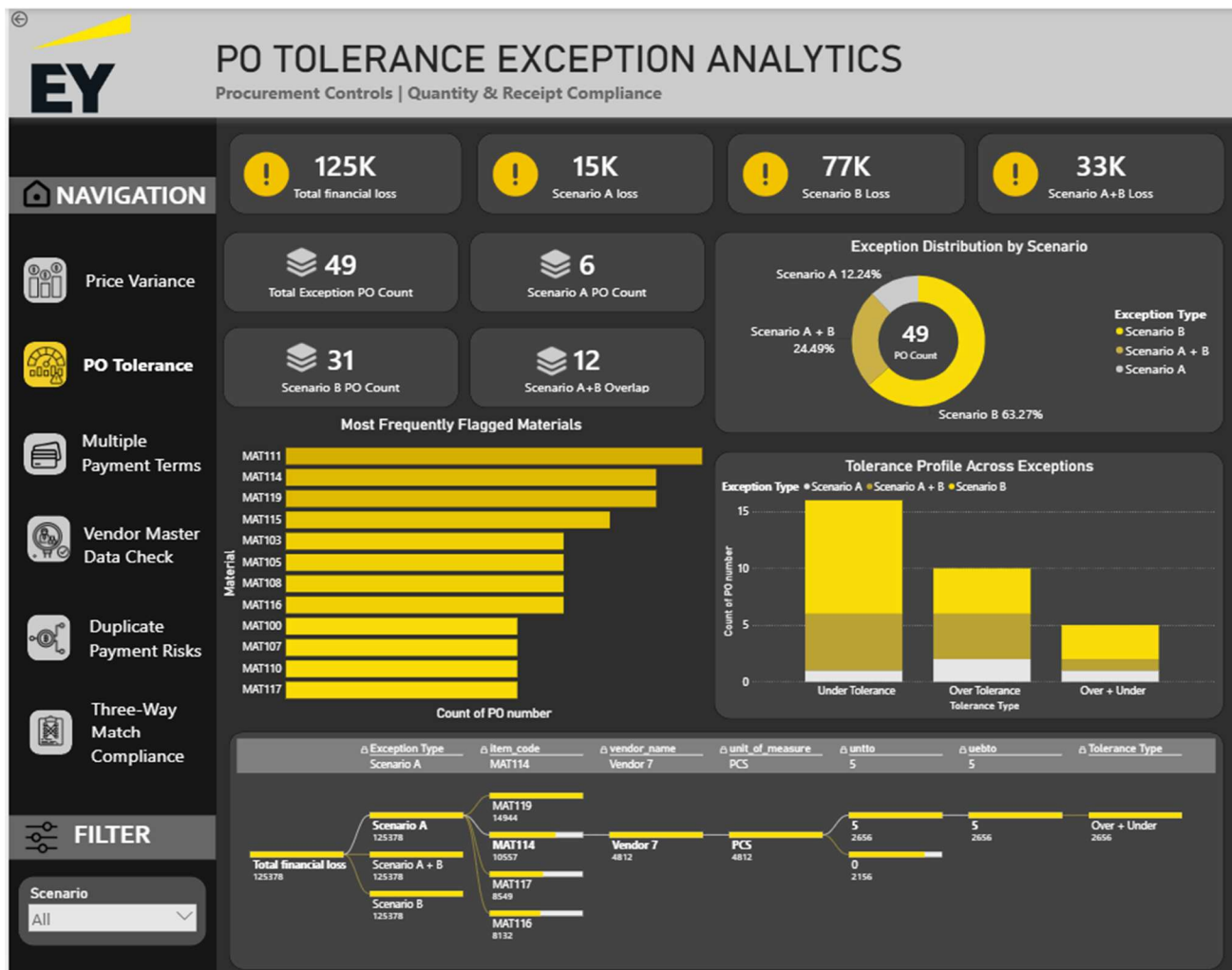
Objective

This scenario evaluates whether **PO tolerance limits** (over-delivery and under-delivery percentages) are configured consistently and appropriately. Scenario A identifies inconsistent tolerance values for the same material and vendor across different purchase orders, while Scenario B detects tolerance settings applied to discrete units of measure such as **Nos.**, **Pcs.**, and **Sets**, where such tolerances are not applicable. The objective is to strengthen quantity control and ensure standardized procurement practices.

Business Risk

Inconsistent or incorrect tolerance settings can result in unauthorized quantity deviations, financial losses due to overpayments or short deliveries, and inconsistent goods receipt practices. They also indicate weak master data governance and increase audit and compliance risks.

Dashboard Overview



PO Tolerance Variance Analysis Dashboard

KPI Explanation

Financial Loss KPIs: The dashboard displays the total estimated financial impact along with scenario-wise losses for Scenario A, Scenario B, and the overlap between both scenarios. The values clearly show that Scenario B contributes the highest financial exposure, indicating that incorrect tolerance configuration for discrete units is a more widespread issue than inconsistent tolerance definitions.

Purchase Order KPIs: These cards show the total number of exception purchase orders and their distribution across each scenario. The overlap count highlights purchase orders that violate both rules simultaneously, helping procurement teams prioritize the highest-risk cases for review.

Dashboard Visual Explanation

Exception Distribution by Scenario (Donut Chart): This visual illustrates how exception purchase orders are distributed across Scenario A, Scenario B, and the overlapping cases. It quickly highlights that Scenario B accounts for most identified exceptions.

Most Frequently Flagged Materials (Bar Chart): The chart ranks materials based on the number of tolerance-related exceptions, helping identify materials that repeatedly appear in configuration issues and should be reviewed first.

Tolerance Profile Across Exceptions (Stacked Bar Chart): This visual compares over-delivery, under-delivery, and combined tolerance exceptions across scenarios, providing insight into which tolerance configuration issues occur most frequently.

PO Tolerance Variance Analysis (Decomposition Tree): The decomposition tree allows users to drill down from exception type to material, vendor, unit of measure, and tolerance values, making it easier to identify the exact configuration responsible for each exception.

Business Insights

The analysis indicates that **incorrect tolerance configuration for discrete units (Scenario B)** is the primary contributor to both exception volume and financial exposure. Standardizing tolerance settings and restricting tolerance percentages for discrete units such as **Nos., Pcs., and Sets** would significantly reduce procurement risk while improving master data quality and consistency across purchase orders.

Possible Real-World Causes

- Tolerance fields left at a legacy default value carried over from a template PO rather than being deliberately configured
- Buyers unaware that tolerance percentages are inapplicable to discrete units of measure
- Material master reclassification from a weight/volume-based unit to a discrete unit without updating the associated tolerance defaults
- Different buyers or plants applying their own informal tolerance conventions in the absence of a documented policy
- Vendor-specific historical exceptions where a wider tolerance was granted for a one-off logistics constraint and never reset

- System configuration inherited from a different plant or company code during a rollout or migration
- Urgent procurement scenarios where tolerance was loosened temporarily to avoid rejecting a time-critical delivery
- Lack of a validation rule in SAP customizing to prevent tolerance entry against discrete UoM materials
- Manual data entry error when replicating a PO template across multiple materials
- Vendor negotiation allowing a wider delivery margin for bulk or continuous-supply materials that was incorrectly copied to discrete materials

Recommended Controls

- Configure SAP validation rules to automatically zero out or block tolerance entry for materials procured in discrete units of measure
- Standardize tolerance percentage bands by material category and enforce them through a reference table rather than manual entry
- Introduce a periodic tolerance-configuration audit report, run monthly, to catch drift before it accumulates
- Require supervisor approval for any PO tolerance value that deviates from the standard band for that material category
- Provide buyer training specifically covering when tolerance is and is not an applicable control

Scenario 3 — Multiple Payment Terms

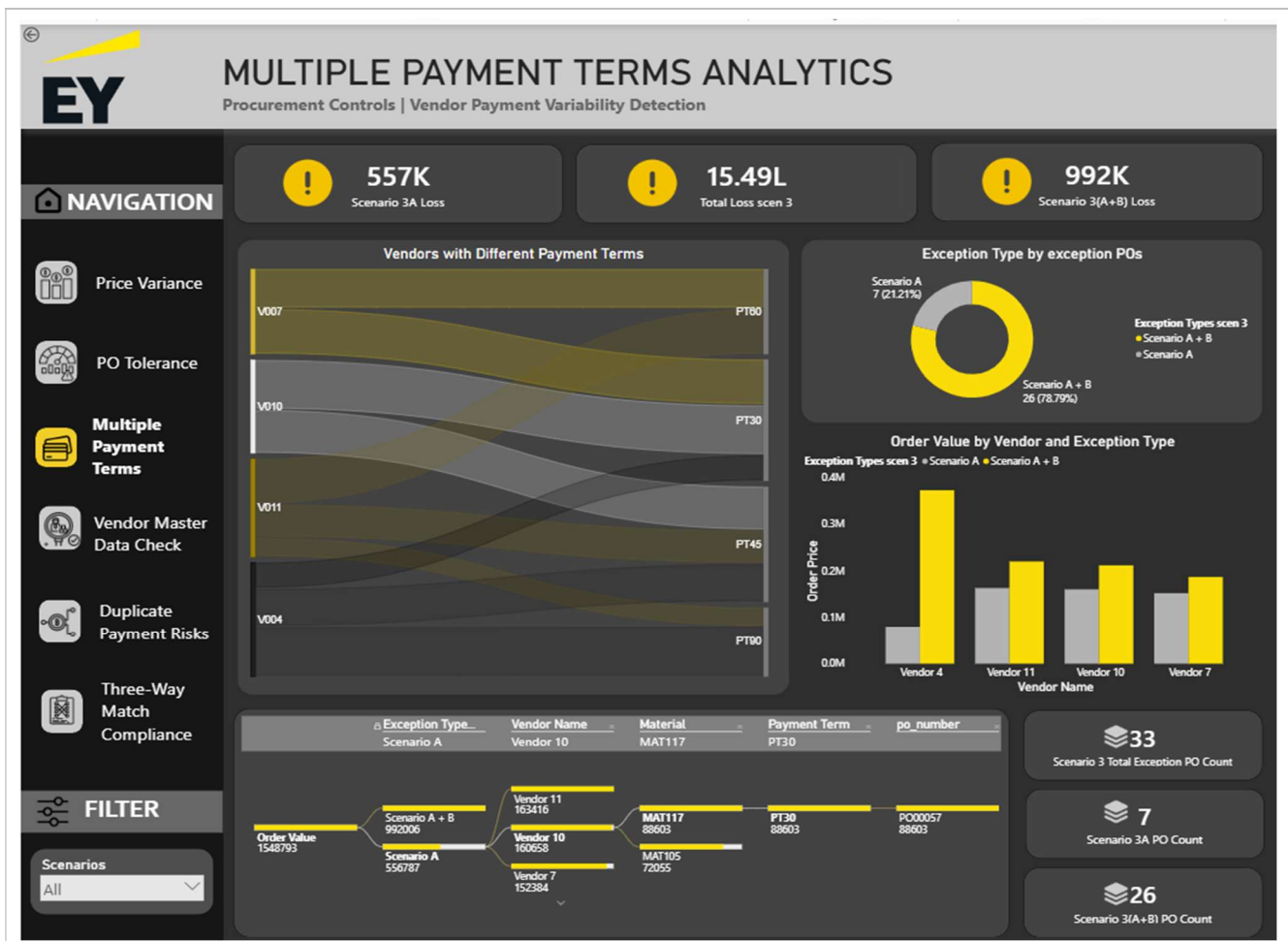
Objective

This scenario evaluates whether a vendor is associated with **multiple payment terms (ZTERM)** across different purchase orders instead of a single agreed commercial term. The objective is to identify inconsistencies in payment conditions that may indicate weak contract governance and to measure the financial exposure associated with these vendors.

Business Risk

Multiple payment terms for the same vendor can lead to unpredictable cash flows, payment processing errors, audit observations, and vendor disputes. It also increases the risk of inconsistent contract application and weak vendor master data governance. Vendor and Business Impact: vendors themselves can become confused about which invoices fall under which term, increasing payment disputes; internally, it also raises the risk of duplicate or accelerated payment if two terms are applied to what is effectively the same commercial obligation.

Dashboard Overview



Multiple Payment Terms Analytics Dashboard

KPI Explanation

Financial Loss KPIs: The dashboard displays the total order value affected by vendors with multiple payment terms, along with the financial impact of Scenario A and the overlapping Scenario A+B exceptions. The values show that the majority of the exposure comes from vendors appearing in the combined exception category.

Purchase Order KPIs: These cards indicate the total number of exception purchase orders and their distribution across each scenario. Most of the affected purchase orders belong to the Scenario A+B category, highlighting the vendors that should be prioritized for payment term standardization.

Dashboard Visual Explanation

Vendors with Different Payment Terms (Sankey Diagram): This visual shows how each flagged vendor is linked to multiple payment terms, making it easy to identify vendors with inconsistent payment conditions across purchase orders.

Exception Distribution by Scenario (Donut Chart): The donut chart displays the proportion of purchase orders falling under Scenario A and Scenario A+B, showing that the majority of exceptions belong to the overlapping category.

Order Value by Vendor (Bar Chart): This chart compares the total order value of vendors with payment term inconsistencies, helping identify vendors contributing the highest financial exposure.

Drill through Detail Table: The drill through table enables users to navigate from summary-level exceptions to individual purchase orders, payment terms, materials, and vendors for detailed investigation.

Business Insights

The analysis shows that payment term inconsistencies are concentrated among a small number of vendors rather than being organization wide. Standardizing payment terms for these vendors would improve cash flow planning, reduce payment disputes, strengthen contract governance, and significantly lower the associated financial exposure.

Possible Real-World Causes

- Vendor renegotiated payment terms mid-relationship but historical open POs were never updated to the new term
- Different plants or business units independently negotiating terms with the same vendor without central visibility
- Urgent procurement requiring an advance-payment term for a specific order while standard orders remained on net terms
- Vendor master payment term field not defaulting correctly into new purchase orders, requiring manual entry that varied by buyer
- Mergers or vendor consolidations where two previously separate vendor codes (with different terms) were mapped to one vendor

- Special project or one-time purchase carrying a negotiated exception term distinct from the vendor's standard terms
- ERP configuration allowing payment term override at the PO header level without approval
- Currency or cross-border payment requiring a different settlement term for a subset of orders
- Human error during PO creation where the wrong payment term code was selected from a similar-looking list
- Legacy contracts with grandfathered terms that were never migrated to the current standard term structure

Recommended Controls

- Lock payment terms at the vendor master level so they default automatically into every PO and cannot be freely overridden at header level
- Require documented approval and a business justification for any PO-level payment term override
- Run a periodic vendor payment-term consistency report to catch drift before it accumulates across multiple POs
- Consolidate duplicate or fragmented vendor codes so payment term history is visible in one place
- Include payment-term consistency as a standing check in vendor onboarding and periodic vendor master review

Scenario 4 — Vendor Master Validation

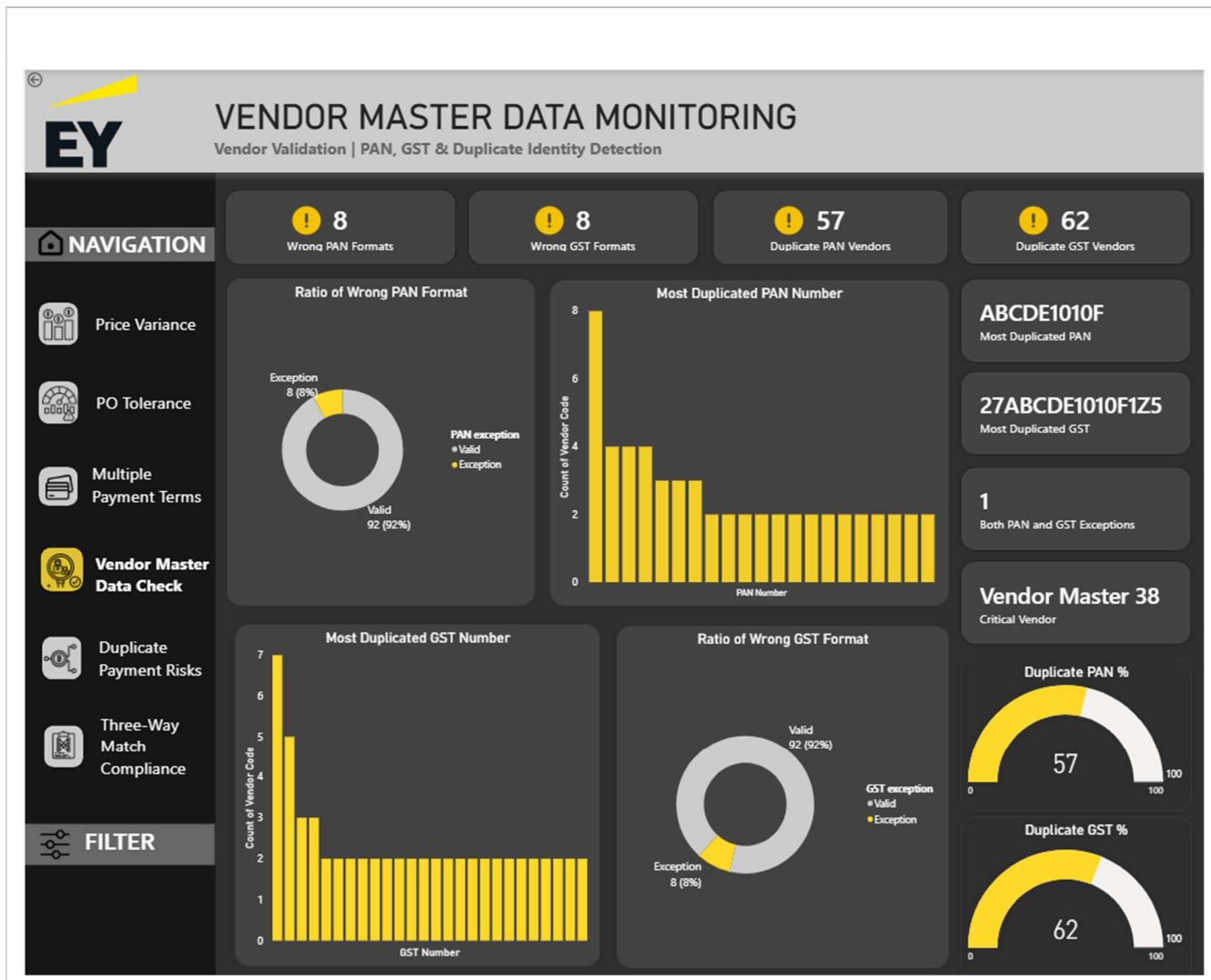
Objective

This scenario validates the quality and integrity of **vendor master data** by checking the format of PAN and GST numbers and identifying duplicate PANs or GST numbers across multiple vendor codes. The objective is to detect master data errors early, strengthen vendor governance, and prevent downstream procurement, compliance, and payment issues.

Business Risk

Invalid or duplicate vendor records can result in duplicate payments, tax compliance issues, inaccurate vendor reporting, and weak master data governance. They also fragment vendor transaction history, making spend analysis, vendor evaluation, and procurement decision-making less reliable.

Dashboard Overview



Vendor Master Data Monitoring Dashboard

KPI Explanation

Invalid PAN & GST Records: These KPI cards show the number of vendors with incorrectly formatted PAN and GST numbers, highlighting basic master data quality issues that require correction before they affect statutory compliance.

Duplicate PAN & GST Vendors: These cards display the number of vendor records sharing the same PAN or GST number, helping identify duplicate vendor creation and potential risks such as duplicate payments and fragmented vendor history.

Most Duplicated PAN & GST: These KPIs identify the PAN and GST numbers linked to the highest number of vendor codes, allowing the master data team to immediately focus on the worst duplicate cases.

Critical Vendor & Combined Exceptions: The dashboard also highlights the vendor with the highest overall risk along with vendors having both PAN and GST issues simultaneously, enabling quick prioritization of master data remediation efforts

Dashboard Visual Explanation

Wrong PAN & GST Format (Donut Charts): These visuals compare valid and invalid PAN/GST records, providing a quick overview of master data quality and showing that format errors are relatively limited.

Most Duplicated PAN & GST (Bar Charts): These charts rank duplicated PAN and GST numbers based on the number of associated vendor codes, helping identify identifiers that require immediate investigation.

Duplicate PAN & GST Percentage (Gauge Charts): The gauges provide an overall indication of duplication levels, allowing management to quickly assess the health of vendor master data

Business Insights

The analysis shows that while **PAN and GST format errors are relatively low**, duplicate PAN and GST records are significantly more common, indicating gaps in vendor onboarding and master data governance. Implementing uniqueness validations during vendor creation would substantially reduce duplicate vendors, strengthen compliance, and improve the reliability of procurement and financial reporting.

Possible Real-World Causes

- Genuinely related legal entities (branches or divisions of the same company) legitimately sharing one GSTIN under GST law while requiring separate vendor codes for different plants
- Vendor onboarded multiple times by different buyers or plants without checking existing master records first
- Legacy vendor codes retained from a system migration alongside newly created codes for the same vendor
- Vendor name changes or rebranding leading to a new code being created instead of updating the existing record
- Manual data entry errors introducing typos into otherwise valid PAN or GST numbers
- Placeholder or test vendor records inadvertently left active in the master data

- Decentralized vendor creation authority across multiple plants without a central deduplication check
- GST number correctly shared across multiple state-registered branches of the same parent vendor (a legitimate GST scenario requiring careful review before flagging as an error)
- Vendor master fields copied from a similar vendor template during rapid onboarding
- Lack of an automated uniqueness validation rule in the vendor creation transaction

Recommended Controls

- Implement a mandatory PAN and GST uniqueness check at the point of vendor creation, with a hard block or required approval on any match to an existing vendor
- Introduce format validation rules in SAP customizing to reject non-conforming PAN and GST entries before the vendor record is saved
- Run a periodic (monthly or quarterly) vendor master deduplication report as a standing control
- Centralize vendor creation approval rather than allowing independent creation across multiple plants
- Require a documented review before merging or deactivating duplicate vendor codes to preserve legitimate transaction history

Scenario 5 — Duplicate Payment Risk

Objective

This scenario identifies **duplicate payment risks** by detecting invoices that have been processed more than once. Scenario A checks for duplicate invoices raised by the same vendor with the same invoice number, date, and amount, while Scenario B identifies invoices with the same date and amount submitted by different vendors. The objective is to prevent financial leakage, strengthen invoice verification controls, and detect potential fraudulent payment patterns.

Business Risk

Duplicate payments result in direct financial losses, increase operational inefficiencies, and weaken invoice processing controls. Cross-vendor duplicates are particularly critical as they may indicate fraudulent activity or collusion, while recovering duplicate payments can be costly and damage vendor relationships.

Dashboard Overview



Duplicate Payment Risk Analytics Dashboard

KPI Explanation

Financial Exposure KPIs: These cards display the total duplicate payment exposure along with the financial impact of same-vendor and different-vendor duplicate payments. The comparison shows that **different-vendor duplicates contribute the larger share of financial risk**, making them a higher investigation priority.

Duplicate Transaction KPI: This card shows the total number of duplicate invoice transactions identified across both scenarios, indicating the overall scale of duplicate payment exceptions requiring review.

Dashboard Visual Explanation

Duplicate Payment Risk Distribution (Donut Chart): This visual compares the proportion of same-vendor and different-vendor duplicate payments, providing a quick understanding of which duplication pattern contributes more to the overall financial exposure.

Vendor-wise Duplicate Payment Exposure (Bar Chart): The chart ranks vendors based on the value of duplicate payments, helping identify vendors with the highest financial exposure and determine whether the issue is concentrated or widespread.

Duplicate Invoice Value Analysis (Column Chart): This visual displays duplicate invoice amounts across invoice numbers, allowing users to identify high-value duplicate transactions that may require immediate attention.

Financial Exposure Trend (Line Chart): The line chart shows how duplicate payment exposure changes over invoice dates, helping identify periods where duplicate payment risk increased.

Drill through Detail Table: Users can drill down from summary-level metrics to individual invoices, vendors, dates, and amounts, enabling detailed investigation and supporting payment recovery activities.

Business Insights

The analysis indicates that duplicate payment exposure is spread across multiple vendors rather than being limited to a single supplier, suggesting a **systemic weakness in invoice verification controls**. The higher financial impact of **different-vendor duplicates** also highlights the need to strengthen duplicate invoice detection mechanisms and investigate potential fraudulent or collusive payment patterns.

Possible Real-World Causes

- Invoice submitted by the vendor twice (once by email, once by portal upload) and both entered into the system without a duplicate check
- Manual re-entry of an invoice after a system error or timeout, without verifying it had already posted successfully
- Invoice processed under two different vendor codes representing the same legal vendor due to a master data duplication issue
- Split invoice processing across two approvers who were unaware the same invoice had already been submitted by the other
- Genuine coincidental match in amount and date between two unrelated vendors, particularly for round-figure or standard recurring charges

- Credit note and original invoice both entered as separate payable line items in error
- System interface or batch upload re-processing the same invoice file a second time
- Vendor consolidation or factoring arrangement legitimately routing one commercial obligation through two vendor codes
- Human error during manual invoice entry duplicating a keystroke sequence for invoice number and amount
- Absence of a system-enforced duplicate invoice number check at the point of posting

Recommended Controls

- Enable a system-enforced duplicate invoice check on vendor, invoice number, date and amount at the point of posting, blocking rather than merely warning
- Extend the duplicate check logic to also screen for cross-vendor matches on amount and date, not only same-vendor matches
- Implement three-way matching consistently before invoice posting to reduce reliance on post-payment duplicate detection
- Run a standing periodic duplicate-payment exception report rather than relying on annual audit sampling alone
- Consolidate duplicate vendor master records to prevent the same legal vendor from bypassing a same-vendor duplicate check

Scenario 6 — Three-Way Match Compliance

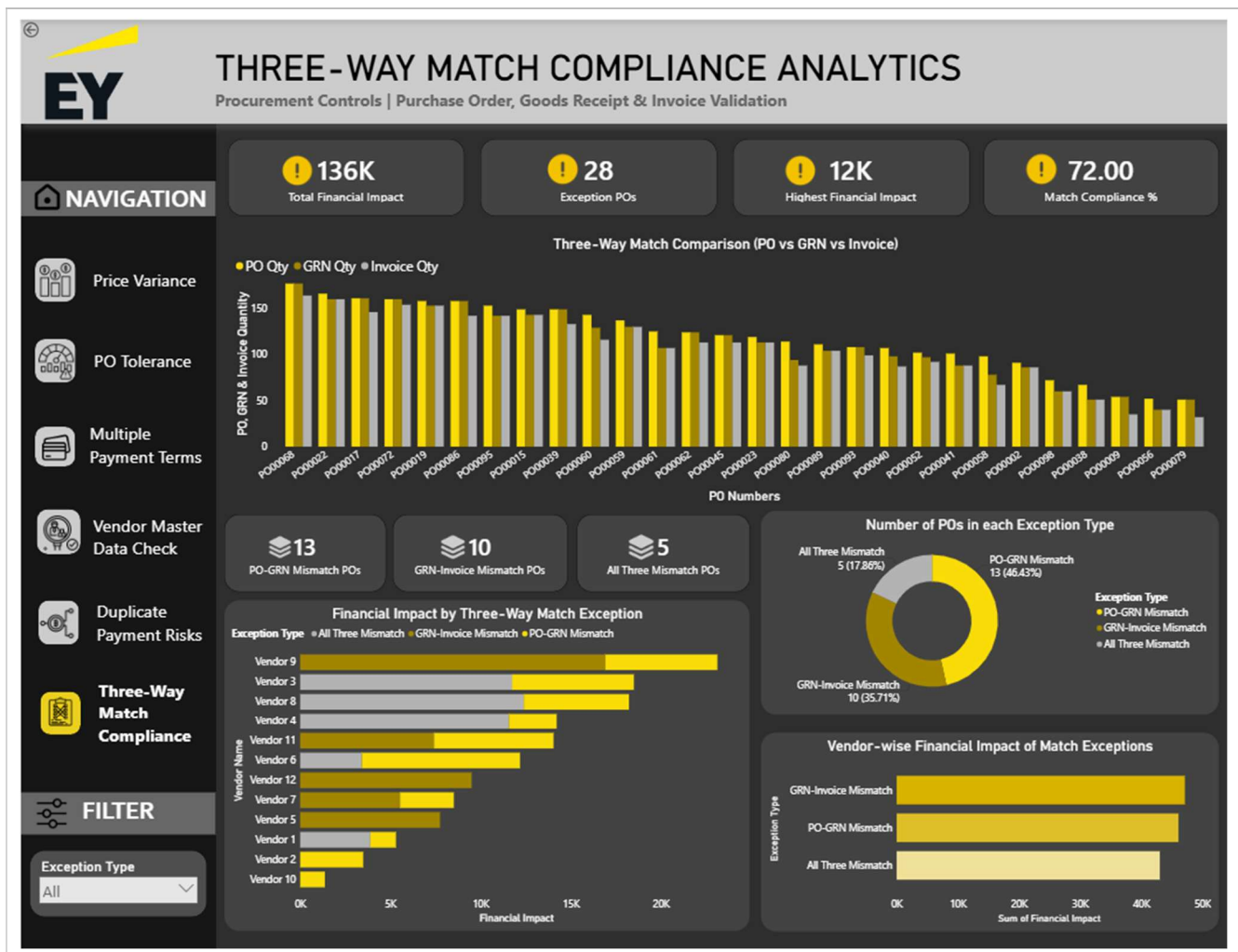
Objective

This scenario evaluates Three-Way Match compliance by comparing **Purchase Order (PO) quantity**, **Goods Receipt (GRN) quantity**, and **Invoice quantity** for every purchase order. Each transaction is classified as a complete match, PO-GRN mismatch, GRN-Invoice mismatch, or All Three Mismatch. The objective is to ensure payments are made only for goods that were ordered, received, and correctly invoiced.

Business Risk

Poor three-way match compliance can result in **overpayments**, **inaccurate inventory records**, **increased manual reconciliation efforts**, and **audit observations**. Frequent mismatches also indicate weaknesses in procurement and invoice verification processes, reducing confidence in vendor performance and internal financial controls.

Dashboard Overview



Three-Way Match Compliance Analytics Dashboard

KPI Explanation

Financial Impact KPIs: These cards display the total financial exposure from quantity mismatches, the highest financial impact observed in a single purchase order, and the overall number of exception purchase orders. Together, they help management understand both the scale and severity of three-way match failures.

Match Compliance KPI: This KPI shows the percentage of purchase orders where **PO Quantity, GRN Quantity, and Invoice Quantity** matched successfully. It serves as the primary indicator of procurement control effectiveness and overall process compliance.

Dashboard Visual Explanation

PO vs GRN vs Invoice Quantity (Clustered Bar Chart): These visual compares ordered, received, and invoiced quantities for each purchase order, making it easy to identify where mismatches occur across the procurement process.

Exception Distribution by Type (Donut Chart): The donut chart displays the proportion of **PO-GRN Mismatch, GRN-Invoice Mismatch, and All Three Mismatch** exceptions, helping identify the most common control failures.

Financial Impact by Vendor (Bar Chart): This chart compares the financial impact of mismatches across vendors, allowing procurement teams to identify vendors associated with the highest exception values.

Financial Impact by Exception Type (Bar Chart): This visual compares the monetary impact of each mismatch category, helping prioritize corrective actions based on financial risk rather than only exception counts.

Business Insights

The analysis shows a **72% Three-Way Match compliance rate**, leaving a significant number of purchase orders with quantity mismatches. Most exceptions occur between the **Purchase Order and Goods Receipt stages**, indicating opportunities to strengthen receiving controls, improve invoice verification, and reduce financial exposure caused by procurement process inconsistencies.

Possible Real-World Causes

- Partial deliveries where the full PO quantity was split across multiple GRNs, only one of which was captured in this analysis window
- Late goods receipt postings where physical delivery occurred but the GRN was not yet recorded at the time of invoice matching
- Vendor shipping a different quantity than ordered due to a stock-availability constraint at their end
- Transportation or logistics damage resulting in a lower accepted quantity than what was originally shipped
- Invoice submitted for the full PO quantity ahead of actual delivery, common in advance-billing arrangements
- Unit of measure conversion errors between the PO, GRN and invoice documents

- Manual GRN entry error where the receiving clerk recorded an incorrect quantity
- Multiple plants receiving against a single centrally raised PO, causing quantity to be split unevenly across GRN postings
- System interface timing issues where a GRN or invoice posting was delayed in reaching the reporting extract
- Return or rejection of a portion of the delivered quantity that was not yet reflected in the GRN at the time of the check

Recommended Controls

- Enforce a system block on invoice posting wherever the invoice quantity exceeds the GRN quantity beyond an approved tolerance
- Introduce a standing weekly three-way match exception report, rather than relying on periodic audit sampling
- Strengthen goods-receipt discipline through mandatory quantity verification at the dock before GRN posting
- Require documented approval for any invoice processed ahead of a corresponding GRN in advance-billing arrangements
- Automate reconciliation alerts for purchase orders showing a mismatch beyond a defined financial threshold

Chapter 5: Dashboard Features

5.1 KPIs and Cards

Every dashboard page opens with a row of card visuals summarizing the most important numbers for that scenario, financial impact, exception counts, and compliance percentages this helps a time-pressed reviewer can absorb the headline story before deciding whether to explore further.

5.2 Filters and Slicers

Each dashboard includes page-level slicers, for example, a Scenario slicer on the PO Tolerance page and an Exception Type slicer on the Three-Way Match page, allowing a user to isolate a specific sub-scenario without navigating away from the page. These slicers interact with every visual and card on the page simultaneously, so narrowing the filter instantly recalculates every KPI in view.

5.3 Tooltips

Several visuals across the scenarios (visible in the file as dedicated tooltip report tabs, use custom tooltip. This allows hovering over a single data point, for instance, one tile in the price variance tree map, to surface a compact, purpose-built mini report with additional context, without requiring the user to leave the main canvas or click through to a separate page.

5.4 Drill through

Drill through is enabled on key fields across the scenarios allowing a user to right-click any summary value and land directly on a filtered detail view scoped to that specific item, vendor or transaction. This effectively gives every summary KPI a one-click audit trail down to transaction level, which is precisely the workflow a risk or audit reviewer needs when investigating a flagged exception.

5.5 Interactive and Cross-Filtering Analysis

All visuals on a given page are cross-filtering by default, clicking a bar, donut slice or tree map tile filters every other visual on the page to that selection. This turns each dashboard from a static report into an exploratory tool.

5.6 Decomposition Trees and Detail Tables

Several scenarios use decomposition tree visuals that let a user progressively expand a summary total across multiple dimensions, Scenario, Material, Plant, Vendor and Date, in whichever order best answers their question, rather than being locked into a single fixed hierarchy. This is paired, in every scenario, with a bottom-of-page detail table that surfaces the specific transactions behind the current filter context.

Chapter 6: Overall Findings

Across the six scenarios, exceptions cluster in identifiable pockets rather than being evenly spread. **Purchase Price Variance** shows the largest exposure (₹13 lakh+), with cross-plant, cross-vendor inconsistency (Scenario D) both most frequent and most costly evidence of weak centralized rate governance. **PO Tolerance** issues stem mainly from one configuration gap (percentages misapplied to discrete units), making this the fastest fix via a single validation rule. **Multiple Payment Terms** (₹15.49 lakh) are concentrated in just four vendors, the narrowest remediation of all. **Vendor Master Validation** finds format entry largely controlled (8% exceptions), but duplicate vendor creation remains a major gap, with 50+ codes implicated per identifier. **Duplicate Payments** (₹8.20 lakh) are driven more by cross-vendor duplication than same-vendor, shifting investigative priority. Finally, **Three-Way Match Compliance** sits at 72%, below benchmark, with most failures at PO-to-GRN pointing to receiving discipline as the key lever for improvement.

A recurring theme across all six scenarios is that a meaningful share of flagged exceptions is, on investigation, likely to have legitimate business explanations, urgent procurement, vendor negotiation outcomes, or system configuration inherited from an earlier process, rather than fraud. The value of this analytics framework is therefore not in presuming wrongdoing, but in surfacing every deviation systematically and consistently, so that the organization can apply informed business judgment to each one rather than relying on ad hoc, sample-based review.

Chapter 7: Recommendations

Based on the findings above, the following overall procurement recommendations are proposed, sequenced roughly by ease of implementation relative to expected control benefit.

- Establish a centralized, system-enforced rate card for high-volume materials, with PO-level price validation against approved rates rather than reliance on post-hoc review
- Configure SAP validation rules to prevent tolerance percentages from being entered against discrete units of measure, directly addressing the largest driver of PO Tolerance exceptions
- Lock payment terms at the vendor master level with controlled override approval, to prevent the same vendor from carrying multiple active terms
- Implement mandatory PAN and GST uniqueness checks at vendor creation, converting vendor master governance from a periodic clean-up exercise into a preventive control
- Extend duplicate invoice checking beyond same-vendor matching to include cross-vendor matching on amount, date and invoice pattern
- Strengthen goods-receipt discipline through mandatory quantity verification at the point of receipt, targeting the PO-GRN mismatch category that drives the majority of three-way match failures
- Establish a standard exception-triage workflow so that every flagged item is reviewed against the "possible real-world causes" checklist before being escalated as a genuine control failure

Chapter 8: Learning Outcomes

This internship gave me hands-on exposure to the complete analytics lifecycle, right from understanding business requirements to building interactive Power BI dashboards.

SAP and Business Logic

I gained a practical understanding of SAP Procure-to-Pay data by working with tables such as **EKKO, EKPO, EKBE, LFA1, LFB1, BSIK and BSAK**. I also learned how business requirements are converted into technical logic using SAP fields and relationships.

Excel and Pivot Tables

Before building any dashboard, I validated every scenario using Excel Pivot Tables. This helped me strengthen my skills in data analysis, grouping, filtering, distinct counts, and validating results before visualization.

Power BI, Power Query and DAX

I developed six complete Power BI dashboards using Power Query for data preparation and DAX for dynamic KPI calculations. I also worked with visuals like decomposition trees, tree maps, Sankey diagrams, donut charts, gauges, drill through pages, tooltips and interactive filters.

Dashboard Design

This project also improved my dashboard design skills. I learned how to present insights in a structured way by selecting appropriate visuals, organizing KPIs logically, and designing dashboards that are easy to understand and interact with.

Risk Analytics and Business Reporting

The biggest takeaway from this internship was learning how to think like a consultant. Instead of just identifying exceptions, I learned how to understand the business risk behind them, estimate their financial impact, analyse possible real-world causes, and present findings in a way that supports better business decisions

Conclusion

This internship gave me hands-on experience of how a risk analytics project is carried out from start to finish. Instead of just building dashboards, I worked on the complete process—understanding the business requirement, preparing a Business Requirement Document (BRD), creating synthetic datasets, validating the logic using Excel Pivot Tables, and finally developing interactive Power BI dashboards.

Across six different Procure-to-Pay risk scenarios, I analysed various procurement and finance risks such as purchase price variance, PO tolerance, payment term inconsistencies, vendor master data quality, duplicate payments, and three-way match compliance. Each scenario was backed by clear business logic, validated through pivot table analysis, and visualized using interactive dashboards to help identify financial impact and operational risks.

Overall, this project helped me understand how data can be used to identify business risks and support better decision-making. It also strengthened my skills in Excel, Power BI, DAX, Power Query, SAP business concepts, and dashboard design. Most importantly, it taught me how to think beyond the numbers by understanding the business reason behind every exception and presenting insights in a way that can help management take informed decisions.

Appendix

A. Business Requirement Documents

B. Pivot Table Validation Workbooks

C. Dashboards